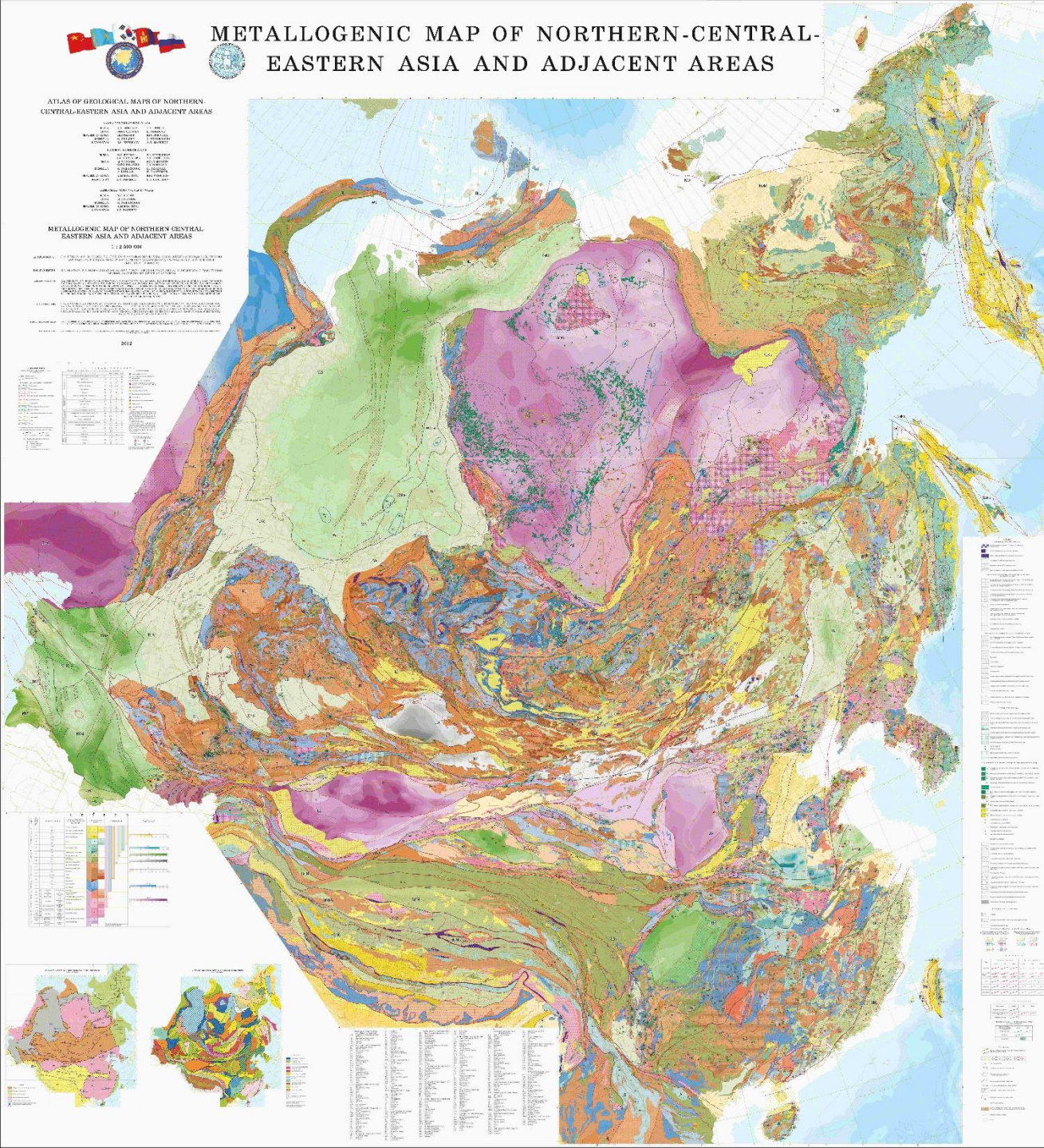


Metallogenic evolution of Northern, Central, and Eastern Asia (based on Metallogenic map of Northern, Central, and Eastern Asia 1:2.5 M with database)

*A.G. Pakhalko, G.A. Shatkov, E.M. Pinsky, Chen Tingyu
G.A.Babin, V.A. Kuznetsov, M.E. Kutyreva*

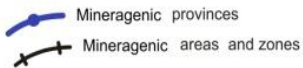


Digital cartography: L. Rucheikova, A. Gureev, M. Demidovich, E. Leontieva, M. Semenova, E. Reneva, M. Shipova, M. Kulchitskaya, V. Guschina, I. Bykova, S. Rebrov, N. Seregina (Россия); Fan Benxian, Ju Yuanjing, Wang Liya, Xie Liangzhen (China) B. Zul, Ts. Minjinsor (Mongolia)

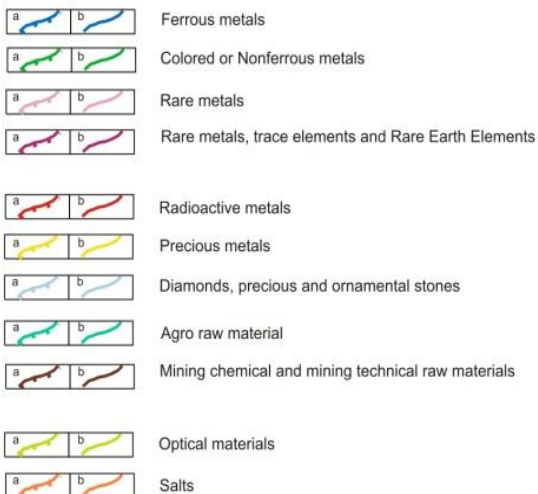
Legend of Metallogenic map

Mineragenic zoning

Boundaries of metallogenic units of different rank metallogenetic



Boundaries of ore zones, regions (a); knots (b)^{1,2}



List of indexes:

1 - The color reflects the metallogenetic specialization, the group of the mineral deposits

2 - Thickness of the lines shows the production of the ore zones, regions and knots:



Metallogenetic unit symbols on the Map

KA - Metallogenetic province

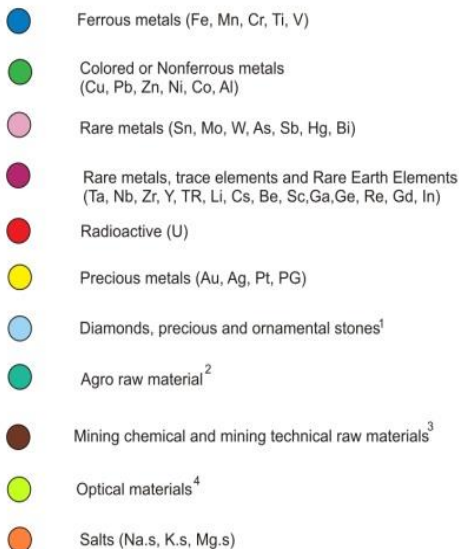
KA-1 - Metallogenetic areas and zones

KA-1.1 - ore zones and regions

KA-1.1.1 - ore knot

KA-1.1.0 - ore knot out of ore zones and regions

Groups of Minerals



List of indexes:

1 - diamond - **di**, amethyst - **amt**, agalmatolite - **agal**, beryl - **be**, turquoise - **tr**, demantoid - **dem**, jadeite - **jad**, emerald - **em**, lazurite - **lz**, nephrite - **np**, piropo - **pir**, rhodonite - **rho**, sillimanite - **sil**, spinel - **sp**, topaz - **to**, tourmaline - **tu**, chalcodony (agate, carnelian) - **ch**, chrysolite - **chy**, chrysopras - **chr**, chrome diopside - **cdi**, charoite - **cha**, jasper - **jas**;

2 - apatite - **ap**, phosphorite - **ph**;

3 - asbestos - **asb**, barite - **ba**, bentonite - **bn**, brucite - **brs**, vermiculite - **ver**, gypsum - **gyp**, graphite - **grf**, kaolin - **kl**, magnesite - **mg**, muscovite - **mus**, opicalcite - **oph**, synnyrite - **syn**, strontianite - **str**, zeolite - **zeo**, talc - **tl**;

4 - iceland spar - **is**, piezooptical quartz - **qz**

Ore quality of deposits (outline colour of the symbol)

	Rich
	Normal
	Poor
	Undefined

Mineral deposits

Genetic types and sizes of deposit marks

Genetic type		Large	Medium	Small	
M a g m a t i c	In layered ultramafic-mafic and alpine-type ultramafic complexes	LUMC			
	Ophiolitic suites	OPH			
	Foidites and foidolites, carbonatites	FAC			
	Kimberlites and lamproites	KIM			
P l u t o g e n i c h y d r o t h e r m a l a n d m e t a s o m a t i c r o c k s	Pegmatite	PEG			
	Greisen and feldspatic in granite and leucogranite	GAF			
	Skarn	SK			
	Telethermal and epithermal	TLT			
	Apogranite	AG			
	Vein and stockwork	VS			
	Porphyry	PORF			
V o l c a n o g e n i c a n d v o l c a n o g e n i c - s e d i m e n t a r y h y d r o t h e r m a l	In volcano-tectonic structure and basement rocks	HPV			
	Volcanogenic and volcanogenic-sedimentary massive sulfide	VMS			
	Stratiform in volcanogenic-sedimentary rocks	SVS			
	Stratiform in carbonate rocks	MVT			
	Sedimentary-exhalation in terrigenous and terrigenous-carbonate rocks	SEDEX			
E x o g e n e t i c - e p i g e n e t i c	Hydrogenous infiltrational and exfiltrational	HIE			
	Sedimentary	SED			
	Placer	PL			
	Waste mantle	WM			
R e g i o n a l o r l o c a l l y t r a n s f o r m e d	Metamorphic	MET			
	Polygenetic	POL			
	Impact	IMP			



ATLAS OF GEOLOGICAL MAPS
OF NORTHERN-CENTRAL-EASTERN
ASIA AND ADJACENT AREAS

METALLOGENY OF NORTHERN, CENTRAL AND EASTERN ASIA

Saint-Petersburg 2018

Explanatory note for Metallogenic map of Northern, Central and Eastern Asia

Editors-in-Chief: O.V. Petrov, A.F. Morozov, E.A. Kiselev, S.P. Shokalsky (Russia), Dong Shuwen, Chen Tingyu (China), O. Chuluun, O. Tomurtogoo (Mongolia), B.S. Uzhkenov, M.A. Sayduakasov (Kazakhstan), Hwang Jae Ha, Kim Bok Chul (Korea)

Authors: G.A. Shatkov, O.V. Petrov, E.M. Pinsky, N.S. Solovyev, V.P. Feoktistov, V.V. Shatov, L.D. Rucheykova, V.A. Gushchina, A.N. Gureev (Russia); Geng Shufang, Chen Binwei, Huang Dianhao, Song Tianrui, Sheng Jifu, Zhu Guanxiang, Sun Guiying, Yan Keming, Min Longrui, Jin Ruogu (China); Dezhidmaa G., Tomurtogoo O. (Mongolia); Bok Chul Kim, Hwang Jae Ha (Republic of Korea); B.S. Uzhkenov, A.L. Kiselev, L.A. Miroshnichenko, D.V. Gurevich (Kazakhstan)

Russian part

Editor: Pakhalko Aleksey

Revised: grammar, structure

uranium deposit. In the Southern Novaya Zemlya area there are also stratiform Paigotinskoe-4 and Razdelnoe Pb-Zn deposits. These deposits **generate-form** the Sakhaminsky ore region. Veinlet-disseminated calcite-fluorite ores of the *Chernogubsky* ore knot, which resources are estimated at 20 million tons of CaF₂, are located in the D₁ limestone.

Fluorite deposits (*Amderma*) are related to the Late Triassic-Early Jurassic (collision) stage of evolution. The issue concerning the age and genesis of the deposit is not understood. Perhaps it formed during **riftogenesis-that-occurred(rifting)** at the early stage of formation of the West Siberian plate (T₁).

In general, lead and zinc ores, fluorite, manganese as well as gold, copper, uranium, **rock crystal**, agate deposits are currently of prime importance in the Paikhoi-Novaya Zemlya province; phosphate and bauxite ores are promising. The Kara and Ust-Kara astroblemes remain poorly understood.

Taimyr-Severnaya Zemlya province (TNL), Russia
565,000 km²

The province is confined to the **self-titled** folded region **of-the-same-name**, which forms part of the Hercynian-Early Cimmerian basement of the Barents-Northern Kara young platform [Proskurnin 2013]. Main minerals are Au, Pb, Zn, Cu, Ni, Mo, U, grf. The province hosts twelve ore deposits and many occurrences, but large ore deposits have not been found yet.

As to the time of consolidation and metallogenic orientation, the province is subdivided into three metallogenic areas: Northern Taimyr-Severnaya Zemlya (TNL-1), Mid-Taimyr (TNL-2) and Southern Taimyr (TNL-3).

Northern Taimyr-Severnaya Zemlya area (TNL-1)

In-the-area-(T)here are five **small** ore deposits, **but-large ore**-deposits have not been found. Main minerals are Au placers (Golysheva River, Tora River, Studenaya River). There **are-no-large ore-knots; they are only** NQ Golyshevsky **medium size** ore **knot-is-of-medium-size**. Source **rocks-for** of gold **are-is** hydrothermally **occurences-inaltered** metamorphosed sedimentary rocks. **High P**otential gold ore knots **have-been-outlined**: Chukchi-Oleninsky (TNL-1.4.0), Konechninsky (TNL-1.6.0), etc.

In the Northern Taimyr-Severnaya Zemlya subprovince, there are two metallogenic zones: Severnaya Zemlya zone (gold-iron-base metals) and Mininsky-Bolshevik zone (gold-rare metals) separated by the Main Severnaya Zemlya Fault.

Chinese part

Editor: Chen Tingyu

Revised: **content (partly and fully rewritten text, updated ages for deposits)**, grammar, structure, references

In the Amu Darya syncline there is a thick halogen strata which is likely to contain specific sodium and other salt deposits. These units are not shown on the Metallogenic Map. There is also information on phosphate occurrences in the south-western part of the area.

Southern Tien Shan metallogenic province (ST), Uzbekistan, Kyrgyzstan, Tajikistan, China
280,000 km²

Southern Tien Shan metallogenic province is an eastern extension of Kyzylkum province structurally located between the Mid-Tien Shan in the north, and Tarim Platform and Paleothetis in the south. The province consists of the Central Asian and Chinese segments with the total latitudinal strike of 2,000 km.

To the south Southern Tien Shan replaces fold structures of the Mid-Tien Shan metallogenic province. Southern Tien Shan differs from the north and mid as it is composed of the Middle Paleozoic rocks formed in geological conditions from mid-ocean ridges to passive continental margin. There are many signs of overthrust-folding tectonic processes and Hercynian orogeny with the tectonic-magmatic activation. There is a difference in the crust structure of the northern part of the province composed of the granite-amphibolite basement, and the southern part composed of the granite-gneiss basement [Kh.N. Baimukhamedov et al., 1985]. The northern part is dominated by clastic sediments; basaltic volcanic rocks and granitoids are widely spread there. In the southern part carbonate rocks and more acidic volcanics and granitoids are widely spread.

Main minerals of the province include Hg, Al, fl, Au, Pb, Sb, Sn, W, U. As opposed to Kyzylkum province the area in question is characterized by abundance of small deposits. There are 233 deposits in the province, 8 are large: *Khaidarkan* (Hg, Sb, fl), ST-2.3.0, Kara-Archa (Hg, Sb), ST-2.3.0, *Dzhizhikrut* (Sb, Hg), ST-1.3.1, *Chonkoy* (Hg), ST-2.5.0, *Kadamjai* (Sb, Hg), ST-2.4.0, *Dzhangu-Bulak* (Hg, Sb, fl, As) ST-2.3.0, *Botzzygoer* (Zr, Nb, TR), ST-4.2, *Fanchankou* (V), ST-5.2.1.

Many ore knots are located within the Late Hercynian ore belt of the northeast strike. Primary mineralization age of the province, except sedimentary deposits of lead and zinc (D₂), bauxite (C₂), copper (K₁₋₂) and strontium (K₂), deposits of Au, Sn, and Sb, Hg, is closely associated with the Permian collisional granitoids. More detailed studies have shown, however, that telethermal Sb and Hg mineralization was formed during the Neogene arch-block activation. Geological and metallogenic peculiarities of the Southern Tien Shan province allow dividing it into five metallogenic areas. They are Zarafshan-Kyrgyz (ST-1), Khaidarkan-Tuyamuyun (ST-2) and Kokshaal (ST-3) areas in the Central Asian segment (ST-1, ST-2, ST-3),

In the Amu Darya, syncline there is a thick halogen strata, which is likely to contain specific sodium and other salt deposits. These units are not shown on the Metallogenic Map. There is also information on phosphate occurrences in the south-western part of the area.

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280,000 km²

The Southern Tien Shan (Tianshan) metallogenic province is the eastern extension of the Kyzylkum metallogenic province. In tectonics, it is sandwiched between the Middle Tien Shan (Tianshan) in the north, and the Tarim block in the south. This metallogenic province consists of some parts in Central Asia and some parts in China with a total length of 2,000 km from west to east in the strike direction.

Different from the Northern Tien Shan (Tianshan) and the Middle Tien Shan (Tianshan), the Southern Tien Shan (Tianshan) consists of Middle Paleozoic rocks formed in geological settings from mid-ocean ridge to passive continental margin. Variscan overthrusting and folding along with the tectono-magmatism were quite severe. There exists a difference in the crustal composition between the northern part and the southern part of this metallogenic province. In the northern part, the basement is composed of granite-amphibolite, while in the southern part, the basement is composed of granite-gneiss [Baimukhamedov et al., 1985, reference?]. The northern part is dominated by clastic sediments; basaltic volcanic rocks and granitoids are widely spread there. In the southern part, carbonate rocks and more acidic volcanic rocks and granitoids are widely distributed

Main minerals of the province include Hg, Al, fl, Au, Pb, Sb, Sn, W, U. As opposed to Kyzylkum province the area in question is characterized by abundance of small deposits. There are 233 deposits in the province, 8 are large: *Khaidarkan* (Hg, Sb, fl), ST-2.3.0, Kara-Archa (Hg, Sb), ST-2.3.0, *Dzhizhikrut* (Sb, Hg), ST-1.3.1, *Chonkoy* (Hg), ST-2.5.0, *Kadamjai* (Sb, Hg), ST-2.4.0, *Dzhangu-Bulak* (Hg, Sb, fl, As) ST-2.3.0, *Botzzygoer* (Zr, Nb, TR), ST-4.2, *Fanchankou* (V), ST-5.2.1.

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Metallogenic Map of Northern, Central, and Eastern Asia

Analysis and compilation of data from [database](#) only

<i>Item No.</i>	<i>Field Description</i>	<i>Field Name</i>
1	Unique ID of the object	ui
2	Deposit name	dn
3	Synonym for the deposit name	sdn
4	Index of deposit on the map	din
5	Country	con
6	Administrative binding	adm
7	Coordinates XX° N, latitude (decimal degrees)	lt
8	Coordinates XX° E, longitude (decimal degrees)	lg
9	Binding to sheets, scale 1:1,000,000	st
10	Group of minerals	gm
11	Main minerals	mm
12	Associated minerals	am
13	Genetic type	gt
14	Geological and commercial type	ct
15	Deposit spread	ds
16	Ore quality	oq
17	Morphology of ore bodies	mor
18	Age of mineralization	age
19	Methods of mining	met
20	Degree of development	sta
21	Ore region, area	are
22	Geological formation conditions	geod

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Ferrous Metals (fm)
<i>Iron (Fe)</i>
<i>Manganese (Mn)</i>
<i>Chrome (Cr)</i>
<i>Titanium (Ti)</i>
<i>Vanadium (V)</i>
Non-Ferrous Metals (nm)
<i>Copper (Cu)</i>
<i>Nickel (Ni), Cobalt (Co)</i>
<i>Lead (Pb), Zinc (Zn)</i>
<i>Aluminium (Al)</i>
Rare Metals (rm)
<i>Tin (Sn)</i>
<i>Molybdenum (Mo)</i>
<i>Tungsten (W)</i>
<i>Antimony (Sb)</i>
<i>Mercury (Hg)</i>
<i>Bismuth (Bi)</i>
<i>Arsenic (As)</i>
Rare, Scattered and Rare Earth Elements (rmm)
<i>Tantalum (Ta), Niobium (Nb)</i>
<i>Lithium (Li)</i>
<i>Cesium (Cs)</i>
<i>Beryllium (Be)</i>
<i>Germanium (Ge)</i>
<i>Rare Earth Elements (TR, Zr, Hf)</i>
Precious Metals (prm)
<i>Gold (Au)</i>
<i>Silver (Ag)</i>
Platinum Group Metals (PG)
Radioactive Elements (rad)
<i>Uranium (U)</i>

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Mining Chemical and Engineering Materials (mtm)
<i>Fluorspar (fl)</i>
<i>Boron (B)</i>
<i>Sulphur (S)</i>
<i>Barite (ba)</i>
<i>Graphite (grf)</i>
<i>Brucite (brs)</i>
<i>Asbestos (asb)</i>
<i>Plaster (gyp)</i>
<i>Muscovite (mus)</i>
<i>Phlogopite (phl)</i>
<i>Vermiculite (ver)</i>
<i>Kaolin (kl)</i>
<i>Bentonite (bn)</i>
<i>Zeolite (zeo)</i>
<i>Magnesite (mg)</i>
<i>Quartzite (qua)</i>
<i>Perlite (per)</i>
<i>Synnyrite (syn)</i>
<i>Talc (tl)</i>
Agricultural Materials (agr)
<i>Apatite (ap)</i>
<i>Phosphorite (ph)</i>
Salts (sal)
<i>Sodium salt (s.Na)</i>
<i>Potassium salt (s.K)</i>
Diamonds, Precious, Semiprecious and Ornamental Stones (dia)
<i>Diamonds (di)</i>
<i>Precious, Semiprecious and Ornamental Stones (dia)</i>
Optical Materials (om)
<i>Optical quartz (gz)</i>
<i>Iceland spar (is)</i>

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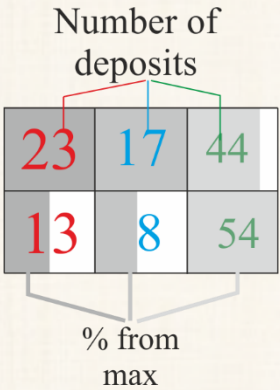
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1	N-Q
2	K ₂ -E
3	J ₃ -K ₁
4	J ₂₋₃
5	PZ ₃ -T ₁
6	PZ ₂
7	PZ ₁
8	NP-Є ₁
9	PP ₄ -MP
10	PP ₁ -PP ₃
11	MA-NA

Metallogenic evolution of Northern, Central, and Eastern Asia

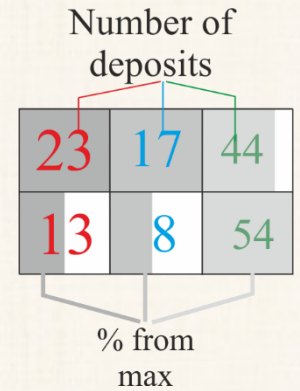
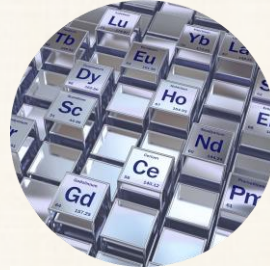
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Ferrous																
Mineral		Fe			Mn			Cr			Ti			V		
Size of deposits		L	M	S	L	M	S	L	M	S	L	M	S	L	M	S
Total number of deposits		90	172	430	7	20	51	1	5	5	15	21	21	9	5	9
1	N-Q		1	5			8		1	1	1	6	4			
2	K ₂ -E	6	4	21		3	2			1	5	10	9			
3	J ₃ -K ₁	5	27	26			1									
4	J ₂₋₃	1	10	26			5				1		2			
5	PZ ₃ -T ₁	11	19	69	1	3	9					1	1			1
6	PZ ₂	18	37	119	1	3	7	1	3	1	1	1	2	5	2	
7	PZ ₁	11	7	48	1	5	9							4	2	8
8	NP-Є ₁	3	17	37	4	6	7		1	2	2	1				
9	PP ₄ -MP	8	4	26			1				2		3		1	
10	PP ₁ -PP ₃	8	9	21			2				1	1				
11	MA-NA	19	38	32							2	1				

Nonferrous																	
Cu			Pb			Zn			Ni			Co			Al		
L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S
48	96	230	29	91	323	13	18	47	5	8	29	8	8	21	29	51	116
		5									2			1			
1	2	8		2	2						8	1	1	4		7	22
5	3	18	5	12	58	2	1	2		1	4		1	2	1	12	18
3	5	6		8	28	1	5	29		4	11	2	4	11		3	1
12	18	64	1	13	80			2	3			3	1	2	10	18	48
22	48	91	15	45	129	8	7	10		2	1	1	1	1	17	9	26
1	14	30	2	6	25	2	3	2								1	
	1	6	2	1	1		1	1	1						1		
	1	3	3	2			1	1			2						1
4	3	2	1	2						1	1					1	
	1								1								

Analysis and compilation of data from database only



Precious and radioactive metals

Mineral		Au			Ag			PGM			U		
Size of deposits		L	M	S	L	M	S	L	M	S	L	M	S
Total number of deposits		231	290	705	12	30	59	5	8	59	46	54	229
1	N-Q	105	93	86		3	1	4	6	58	1	6	14
2	K ₂ -E	49	9	19		2	3				23	17	44
3	J ₃ -K ₁	25	63	182	7	19	34				13	8	54
4	J ₂₋₃	7	27	70	3	2	7				1	3	12
5	PZ ₃ -T ₁	10	32	142	1	1	10	9					20
6	PZ ₂	18	22	111	1	2	4		7		6	16	47
7	PZ ₁	5	28	59							2	2	7
8	NP-Є ₁	10	10	24					1			2	14
9	PP ₄ -MP		2	11									6
10	PP ₁ -PP ₃	1	2	1		1							10
11	MA-NA	1	2							1			1

Rare metals

[illegible]

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis and compilation of data from [database](#) only



Magmatic

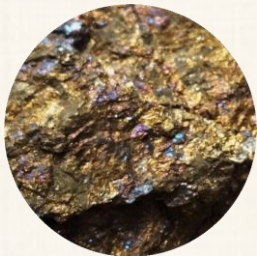


Hydrothermal and metasomatic rocks

Genetic type		LUMC			OPH			FAC			KIM			PEG			GAF			SK			TLT			AG			VS			PORF		
Size of deposits		L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S			
Total number of deposits		35	26	48	21	19	27	39	21	40	15	4	4	38	32	78	40	54	170	59	150	365	19	38	224	10	4	7	124	259	952	49	48	112
1	N-Q																					2							2	2	2			1
2	K ₂ -E		2	1		1	2							1	1		1		3	2		3		2	2				5	10	30	1		4
3	J ₃ -K ₁			3		1	4		1	2				2	2	20	11	14	51	14	37	46	5	7	43	2		1	49	95	392	20	6	26
4	J ₂₋₃		1	2	1			5						4	2	2	11	12	35	5	18	19	5	6	17	3	1	3	6	42	126	4	7	6
5	PZ ₃ -T ₁	8	3	4	5	7	10	3	4	17				3	9	19	11	16	58	16	38	144	8	19	134	1		1	17	28	136	8	16	33
6	PZ ₂	8	12	11	7	4	2	16	6	3	13	4	4	9	4	3	4	12	19	15	38	87	1	4	22	4	3	2	26	40	173	11	17	36
7	PZ ₁	7		13	4		1	2	1	5	1			2	1	1	2		4		14	52			2				3	23	53	3	2	6
8	NP-Є ₁	3	4	7	2	3	5	3	2	3				11	4	4				2	2	3			2				10	14	29			
9	PP ₄ -MP	3	1	4	1	2	3	6	5	7	1			1	1	5				2		3							4	1	10			
10	PP ₁ -PP ₃	4	3	2				3	2	3				5	8	21				3	3	7							1	2	1	2		
11	MA-NA	2		1	1	1		1								3						1							1	2				

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis and compilation of data from [database](#) only



Volcanogenic and volcanogenic-sedimentary



Exogenic Epigenetic

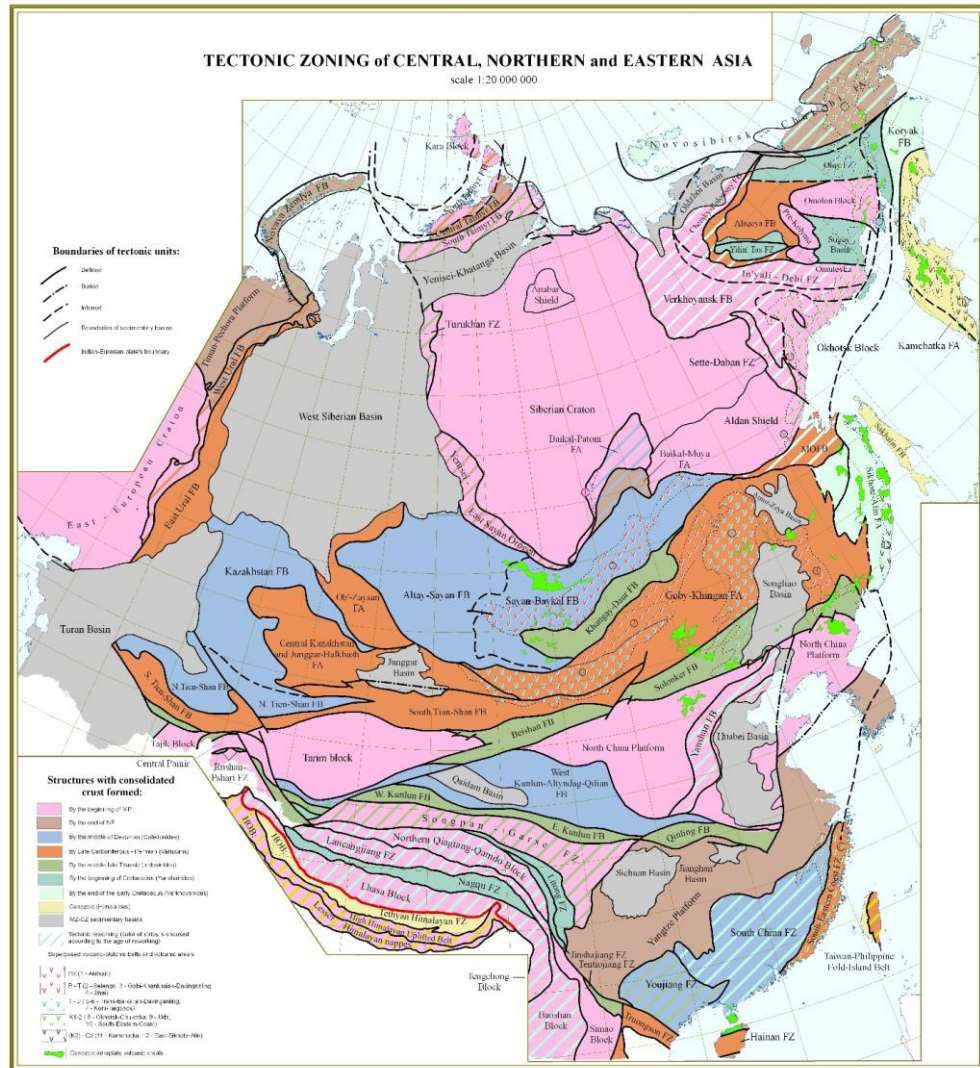


Regionally or locally transformed

Genetic type		HPV			VMS			SVS			MVT			SEDEX			HIE			SED			PL			WM			MET			POL			IMP		
Size of deposits		L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S			
Total number of deposits		53	102	292	33	67	125	28	67	103	10	19	51	8	6	5	25	21	74	224	209	448	181	154	273	23	41	117	79	128	182	27	35	65	2		
1	N-Q	1	6	8			1		3								4	13	11	14	25	127	142	255	3	7	21		1			1					
2	K ₂ -E	2	9	26			2		1	1		1	1	1	3	2	22	13	39	17	27	55	51	11	11	3	10	34	1	1			1	1	2		
3	J ₃ -K ₁	21	46	151	1		3	8	12	5				1				1	7	22	22	72	1		2	2	5	12	1			1					
4	J ₂₋₃	13	24	44		2	6		3	8		3	7				1	1	5	5	5	19	1		3	12	17	40		1	6	1	1				
5	PZ ₃ -T ₁	13	7	34		2	3	2	5	9		2	11							36	32	74		1	1	1	1	5	5	1	6	5	14	36			
6	PZ ₂	3	7	28	26	48	73	12	24	48	6	11	31	4	3	1	1	2	8	19	19	77			1		1	3	8	22	9	1	2	7			
7	PZ ₁		3	1	4	12	31	4	11	18						2	1		1	78	43	76				1		1	4	13	3	2	3	1			
8	NP-Є ₁					2	5	2	1	2	3	1								34	37	36	1			1		1	8	7	19	5	5	13			
9	PP ₄ -MP				1		1		5	9				1						1	5	12							10	10	32			2			
10	PP ₁ -PP ₃				1				1	2	1	1	1	1						1	5	2							14	27	37	11	6	5			
11	MA-NA					1			1	1								1											28	45	70	1	2				

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis of **spatial data** and **database**



Structures with consolidated crust formed:



Superposed volcano-plutonic belts and volcanic areals



Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis of spatial data and database

Number and specialization of **large** deposits within structures of different age of consolidated crust

1	By the beginning of MP
2	By the end of NP
3	By the middle of Devonian (Caledonides)
4	By Late Carboniferous - Permian (Variscans)
5	By the middle-late Triassic (Indosinides)
6	By the beginning of Cretaceous (Yanshanides)
7	By the end of the Early Cretaceous (Verkhoyanides)
8	Cenozoic (Himalaides)

Epochs	Total	LUMC	mm	OPH	mm	FAC	mm	KIM	mm	PEG	mm	GAF	mm	SK	mm	TLT	mm	AG	mm	VS	mm	PORF	mm	HPV	mm	VMS	mm
8	3																							1	S, Cu, Au		
7	16			1	Ti							2	Sn, W	3	W, B, Pb, Zn	2	Hg, W			5	Sn!, Au					1	Pb, Zn
6	45	2	Fe, Ti, ap			2	Y, Ce, ap, Fe					2	Sn, W	1	Sn, Cu, W					18	Sn!, Au, W	2	Cu, Sn, Mo, W	5	Au, Ag		
5	28	1	Cu, Co	1	asb	1	TR, Ce,			4	qz, Li, Sn, Be	1	W	1	Fe, Mo			1	Ta, Nb	1	Au	1	Mo	1	fl		
4	261	5	Ni, Co, Ti	10	asb, V, Ti, Cr,	4	TR, Ce, fl,			2	Ta, Nb	18	W, Mo, Be, fl	11	Fe, W, Cu, Au	9	Au, Hg, Sb	5	Ta, Nb, Zr	21	Au, Be, qz, Sn	11	Cu, Mo,	25	fl, U, Mo,	23	Cu, Pb, Zn
3	243	9	Fe, Ti, Ni, Co	8	asb, jad	9	Al, ap, TR			12	mus, Be, Li	12	W, Mo, Be	22	Fe, W, Au, Cu	5	Au, Ag, Al, fl	4	Zr, Y, Li, Ta,	23	Au!, U, Co, Ni	19	Mo, Cu,	6	zeo, Au,	6	Pb, Zn, Au
2	122	4	Fe, Ti, V	1	asb	6	Al, TR			1	Li			1	Cu	9	Hg!, Sb	1	Ta, Nb	14	Sn, Au, qz, Mo	3	Cu, Fe, Mo	2	Au	1	Cu, Fe
1	443	18	Cr, Ni, Co, Ti, V	2	Cr, tlk	17	Al, ap, Ta, Nb	12	Diam	18	mus, Ta, Nb	4	Sn, TR, W	21	Fe, Mo, Au	2	Qz	1	Ta, Nb	44	Au!, U!, Pb,	16	Mo, Cu	17	is, Au	2	Cu, Pb, Zn

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis of spatial data and database

Number and specialization of **large** deposits within structures of different age of consolidated crust

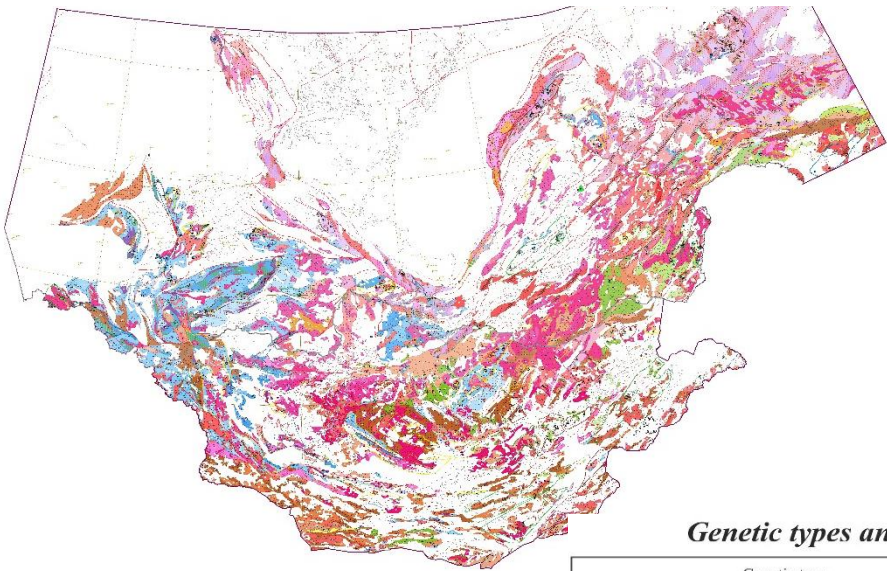
- 1 By the beginning of MP
- 2 By the end of NP
- 3 By the middle of Devonian (Caledonides)
- 4 By Late Carboniferous - Permian (Variscans)
- 5 By the middle-late Triassic (Indosinides)
- 6 By the beginning of Cretaceous (Yanshanides)
- 7 By the end of the Early Cretaceous (Verkhoyanides)
- 8 Cenozoic (Himalaides)

Epochs	Total	SVS	mm	MVT	mm	SEDEX	mm	HIE	mm	SED	mm	PL	mm	WM	mm	MET	mm	POL	mm	IMP	mm
8	3									1	Ge	1	Pt								
7	16											2	Au, Pt								
6	45	1	Fe, Mn							3	Cu, B, gyp	9	Au!								
5	28									2	Cu, V	11	Au!, Sn			2	Ti, V, Fe, ap				
4	261	8	Pb, Zn, Mo, jas			1	Fe	13	U!, Se	22	Al, Fe, Ge, gyp, kl, U	48	Au!, Ti, Zr	5	Co, Ni, kl, Al	14	grph, asb	5	Au!		
3	243	13	Pb, Zn, Fe, Mn	7	Pb, Zn, ba	4	Zn, Pb, Sn, ba	12	U!, Se	23	s. Na,K, Fe, Al, ph,	32	Au!, Ti, Zr, TR,	10	kl, Co, Ti, TR	5	Fe, asb, grph	2	Au, W		
2	122	2	Fe, Mn	3	Pb, Zn, fl					59	ph!, Cu, Fe, Al	4	TR, Sn	3	kl	4	ap, Mn, phl	1	Au		
1	443	8	bn, Cu, Fe, Mn	2	Pb, Zn	3	B, ba, Pb			101	s. Na,K, Fe, Al, ph,	73	Au!, Ti, Zr			54	Fe, grph	18	mg	2	Diam

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis using **Tectono-Metallogenic settings**

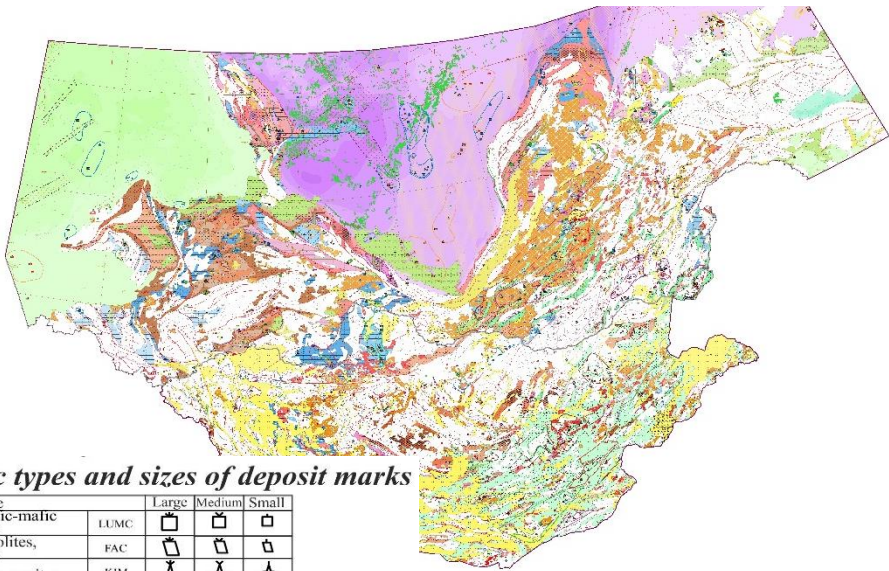
1. Map of **Accretional-Collisional Tectonics** and Related Mineral Deposits



Genetic types and sizes of deposit marks

Genetic type			Large	Medium	Small
Magmatic	Ophiolitic suites	OPH	⬮	⬮	⬮
	Pegmatite	PEG	⬮	⬮	⬮
Plutogenic hydrothermal and metamorphic rocks	Greisen and feldspatic in granite and leucogranite	GAF	⬮	⬮	⬮
	Skarn	SK	⬮	⬮	⬮
	Vein and stockwork	VS	⬮	⬮	⬮
	Porphyry	PORF	⬮	⬮	⬮
Volcanogenic and volcanogenic-sedimentary hydrothermal	In volcano-tectonic structure of central type and diatreme	HPV	⬮	⬮	⬮
	Volcanogenic and volcanogenic-sedimentary massive sulfide	VMS	⬮	⬮	⬮
	Stratiform in volcanogenic-sedimentary rocks	SVS	⬮	⬮	⬮
Exogenetic-epigenetic	Placer	PL	⬮	⬮	⬮
	Waste mantle	WM	⬮	⬮	⬮
Regional or locally transformed	Metamorphic	MET	⬮	⬮	⬮
	Polygenetic	POL	⬮	⬮	⬮

2. Map of **Intraplate Tectonics, Magmatism** and Related Mineral Deposits



Genetic types and sizes of deposit marks

	Genetic type		Large	Medium	Small
			⬮	⬮	⬮
Magmatic	In layered ultramafic-mafic complexes	LUMC	⬮	⬮	⬮
	Foidites and foidolites, carbonatites	FAC	⬮	⬮	⬮
	Kimberlites and lamproites	KIM	⬮	⬮	⬮
	Rare metall alkaline granite	AG	⬮	⬮	⬮
Plutogenic hydrothermal and metamorphic rocks	Pegmatite	PEG	⬮	⬮	⬮
	Greisen and feldspatic in granite and leucogranite	GAF	⬮	⬮	⬮
	Skarn	SK	⬮	⬮	⬮
	Telethermal and epithermal	TLT	⬮	⬮	⬮
	Vein and stockwork	VS	⬮	⬮	⬮
	Porphyry	PORF	⬮	⬮	⬮
Volcanogenic and volcanogenic-sedimentary hydrothermal	In volcano-tectonic structure of central type and diatreme	HPV	⬮	⬮	⬮
	Stratiform in volcanogenic-sedimentary rocks	SVS	⬮	⬮	⬮
	Stratiform in carbonate rocks	MVT	⬮	⬮	⬮
	Sedimentary-exhalation in terrigenous and terrigenous-carbonate rocks	SEDEX	⬮	⬮	⬮
Exogenetic-epigenetic	Hydrogenous infiltrational and exfiltrational	HIE	⬮	⬮	⬮
	Sedimentary	SED	⬮	⬮	⬮
	Placer	PL	⬮	⬮	⬮
	Waste mantle	WM	⬮	⬮	⬮
Regional or locally transformed	Metamorphic	MET	⬮	⬮	⬮
	Polygenetic	POL	⬮	⬮	⬮
	Impact	IMP	⬮	⬮	⬮

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis using **Tectono-Metallogenic settings**

Deposits of different tectonic settings

Intraplate Tectonics and Magmatism					
LUMC	FAC	KIM	TLT	MVT	SEDEX

Accretion and Collision Structures						
OPH	GAF	SK	PORF	HPV	VMS	MET

	By the beginning of MP
	By the end of NP
	By the middle of Devonian (Caledonides)
	By Late Carboniferous - Permian (Variscans)
	By the middle-late Triassic (Indosinides)
	By the beginning of Cretaceous (Yanshanides)
	By the end of the Early Cretaceous (Verkhoyanides)
	Cenozoic (Himalaides)

		Accretion and Collision Structures	Intraplate Tectonics and Magmatism
1	N-Q	Au, Cu 2	
2	K ₂ -E	Cu, Mo, Au, Fe 7	1 Pb, Zn
3	J ₃ -K ₁	Mo, W, Au, Ag, Sn, U 69	6 Au, Fe
4	J ₂₋₃	Mo, W, fl, Cu, Au 34	11 TR, Au, Ag, fl
5	PZ ₃ -T ₁	W, Mo, Fe, grph, Cu 58	21 Co, Ni, Cu, Hg, Pb, Zn, Ti
6	PZ ₂	Cu, Fe, Pb, Zn, Au, Mo 74	48 Diam, Al, Pb, Zn, Fe
7	PZ ₁	Au, Zn, Mo, asb 17	10 Cr, Ti, Fe, diam, Zr, Ta
8	NP-Є ₁	Fe, grph, asb 12	9 Ti, Pb, Ta, Nb
9	PP ₄ -MP	Fe, apat 14	12 Apat, Pb, Zn, Ti, Ni, Cu
10	PP ₁ -PP ₃	grph, Fe, Cu, Pb, Zn 21	9 Ti, Fe, apat
11	MA-NA	Fe, grph, phl 28	3 Ni, Cu, Ti, apat

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis using **Tectono-Metallogenic settings**

Gold deposits of different tectonic settings

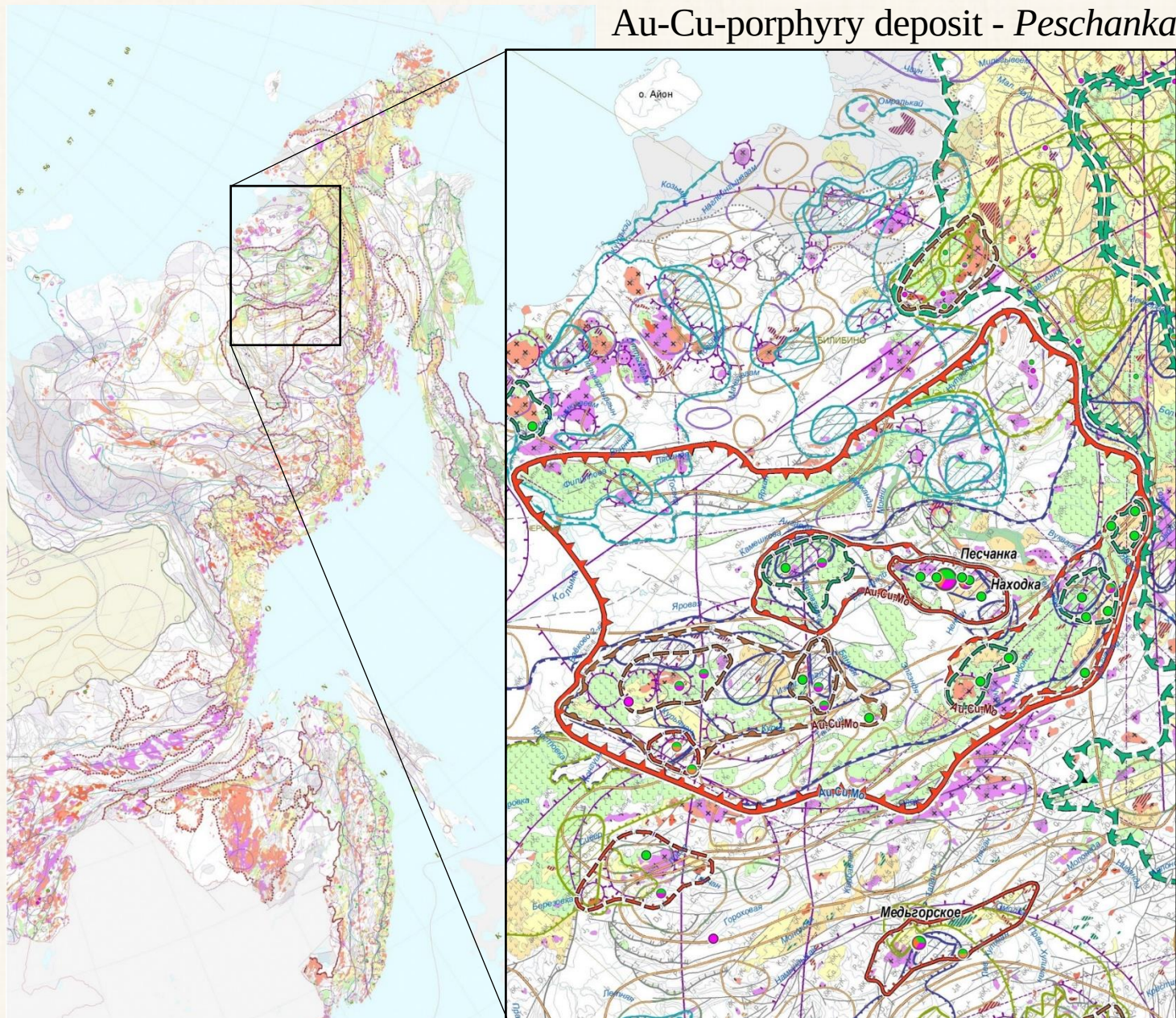
Geological and commercial, and mineragenic types of deposits and generalized groups

		Precious Metals (prm)
		Gold (Au)
	Au-1	Au-quartz-adular
	Au-3	Au-quartz-low-sulphide
	Au-4	Au-sulphide
	Au-5	Au-black shale
	Au-6	Au-Ag
	Au-7	Au-porphyritic
	Au-8	Au-skarn
	Au-2	Au-quartz
	Au-14	Gold complex ores deposits (sulphide, Cu-Ni, U-Au-polymetallic, Cu-Mo-porphyritic, etc.)

Number of
deposits
HPV (269)

		Accretion and Collision Structures	Intraplate Tectonics and Magmatism
1	N-Q		HPV (161) 10 VS (11)
2	K ₂ -E	VS (3) PEG (1) 4	HPV (135) TLT, VMS (2) 26 VS (258)
3	J ₃ -K ₁	HPV (144) TLT (1) SK (232) VS (11350) PORF (11) 80	HPV (21116) TLT (4513) 212 POL (1) VS (1736107)
4	J ₂₋₃	HPV (269) VS (1517) SK, PORF (13) SVS (1) 45	HPV (1) TLT (13) 49 VS (21329)
5	PZ ₃ -T ₁	POL (416) SK (523) VS (21) PORF (22) 55	TLT (4223) 129 POL (5620) VS (41057)
6	PZ ₂	HPV (1) TLT, SVS, VMS (222) POL (125) SK (218) VS (1114) PORF (235) 52	HPV (16) TLT (14) 106 VS (111464)
7	PZ ₁	HPV (1) VS (312) SK (44) 24	HPV (1) 68 VS (21630)
8	NP-Є ₁	VS (4) 4	39 VS (9919)
9	PP ₄ -MP	VS, PEG, SVS, SK (4) 4	9 VS (18)
10	PP ₁ -PP ₃	SVS (1) 1	3 VS (12)
11	MA-NA		3 VS (12)

Au-Cu-porphyry deposit - *Peschanka*



Other criteria

Morphostructures criteria

Ring structures

Tectonic

- Arch, high, dome

Plutonic

- Outcrop
- Buried

Volcanic

- Negative landform
- Positive landform

Faults

- Regional faults
- Planetary faults

Geophysical criteria

Prospect areas

Areas no prospects

- differentiated, alternating magnetic and gravity fields
- poorly differentiated gravity field
- poorly differentiated magnetic field

Tectonic criteria

Faults

Boundaries

- Steeply-dippings and complicated morphology
- Geological boundaries
- Boundaries between the even-aged units of the different facies
- Blind
- Overthrust

East-European platform & Siberian craton

Geochemical criteria

Boundaries of geochemical units

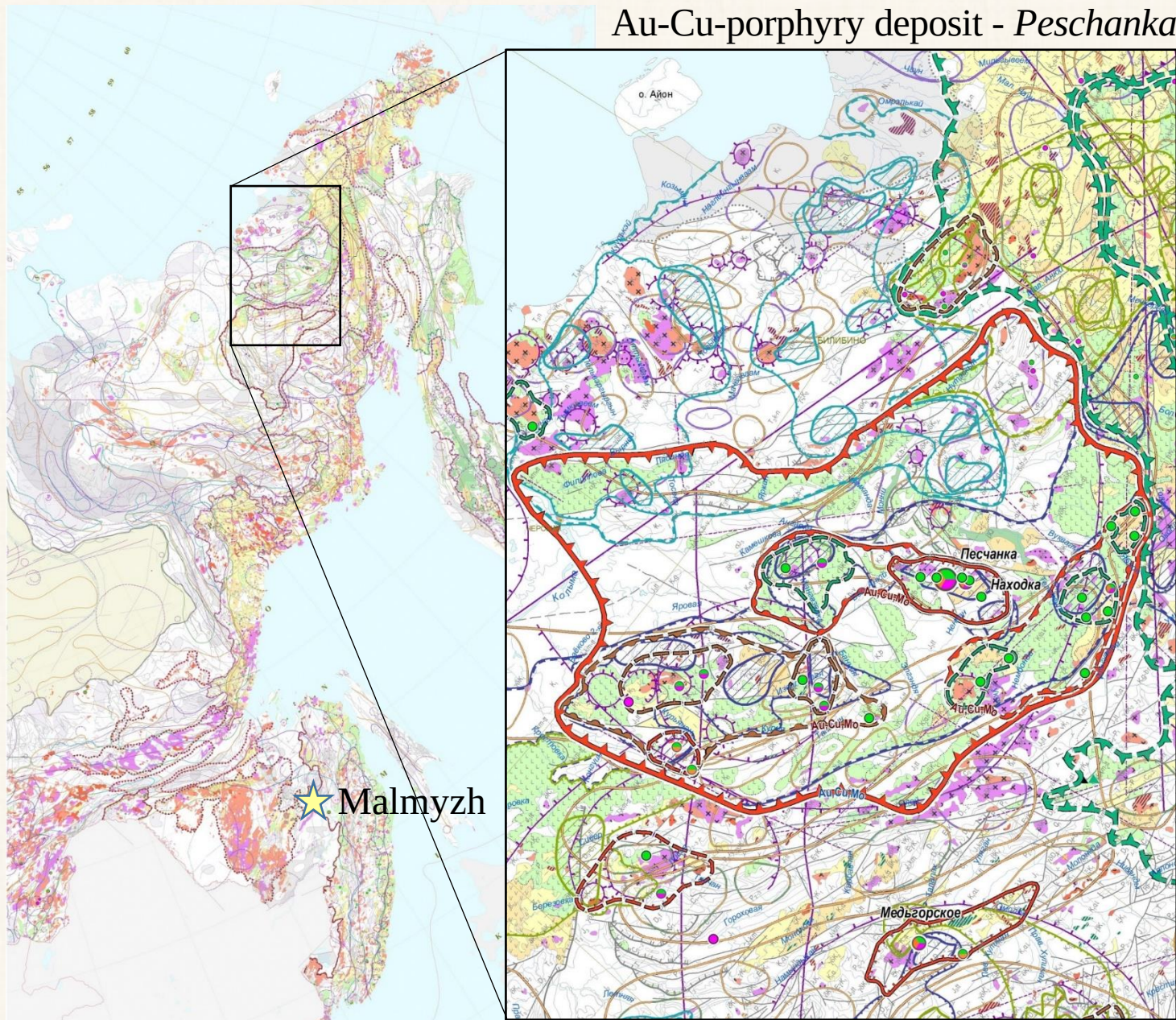
Zones

Regions

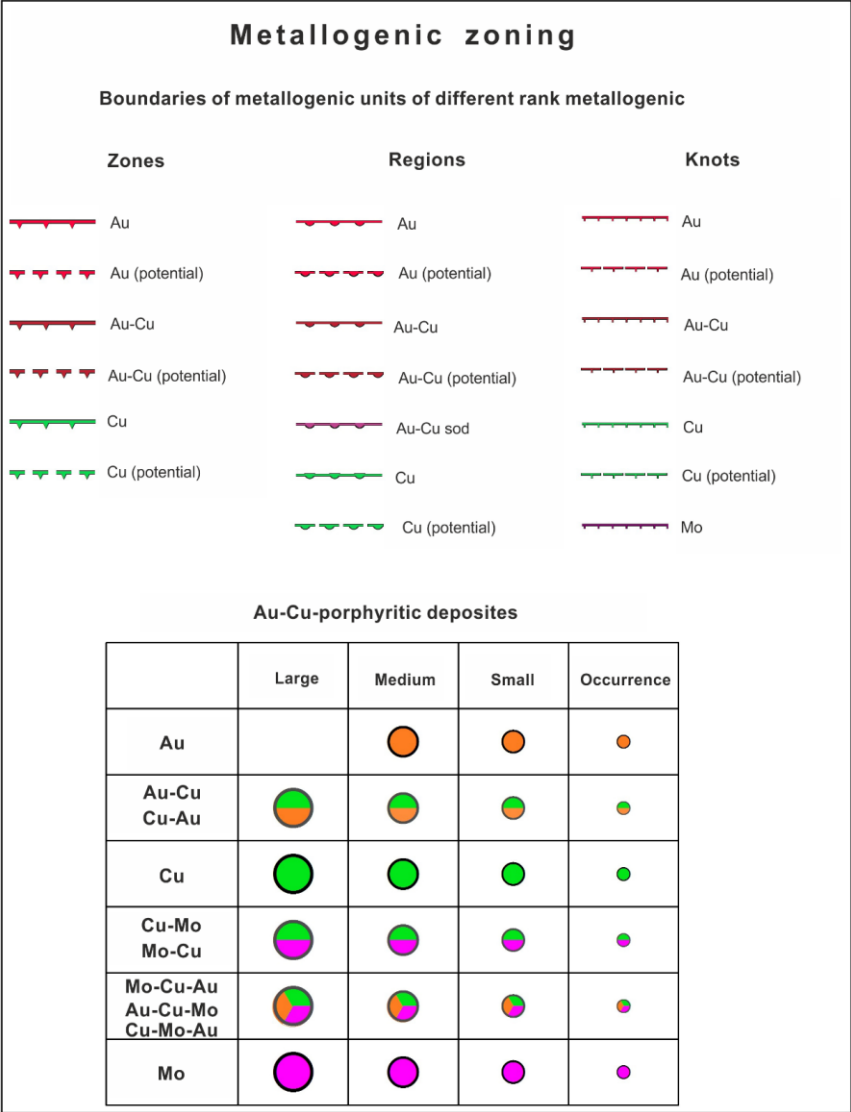
- precious metals
- colored or nonferrous metals
- rare metals

Metasomatic criteria					
Ore-bearing hydrothermally-metasomatic formations					
Conditions of metasomatic forming	Petrologic speciality	Rank of areas			
		areas	regions	knots	sites
PLUTONIC	OKC - Oktaletallic (orthoclase-carbonate-sericite-tourmaline-albite) indifferent, borealite-oktaletallic, biotite-oktaletallic, gumbelie-oktaletallic (tservenite-oktaletallic, propylite-oktaletallic)				
	Фил - Phyllitic type; phyllitic zone indifferent, gumbelie-phillitic				
	ГББ - Gumbelie-beresilic type				
VOLCANOGENIC	КФ - K-spar-porphyrific type				
	ФФ - Feldspathophyrific type				
	АФ - Albitophyrific type				

Au-Cu-porphyry deposit - Peschanka

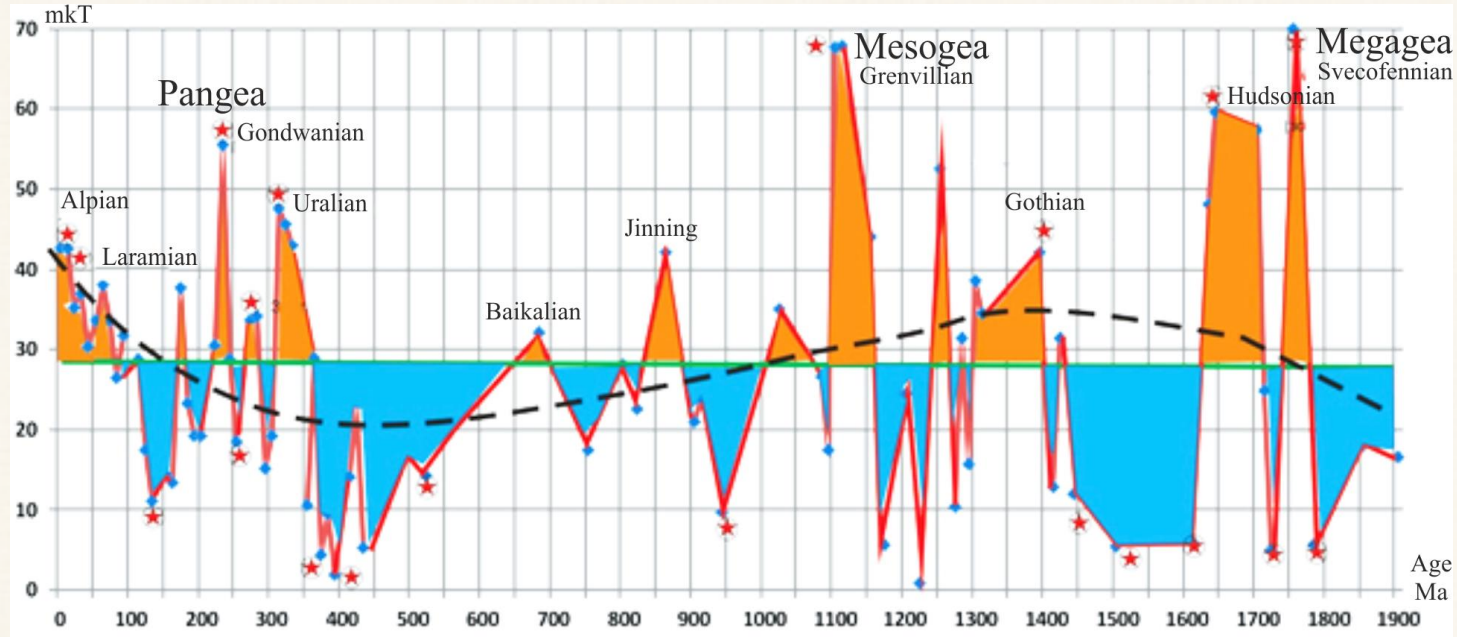


Prognosis



Variations of **Intensity of Earth's magnetic field**

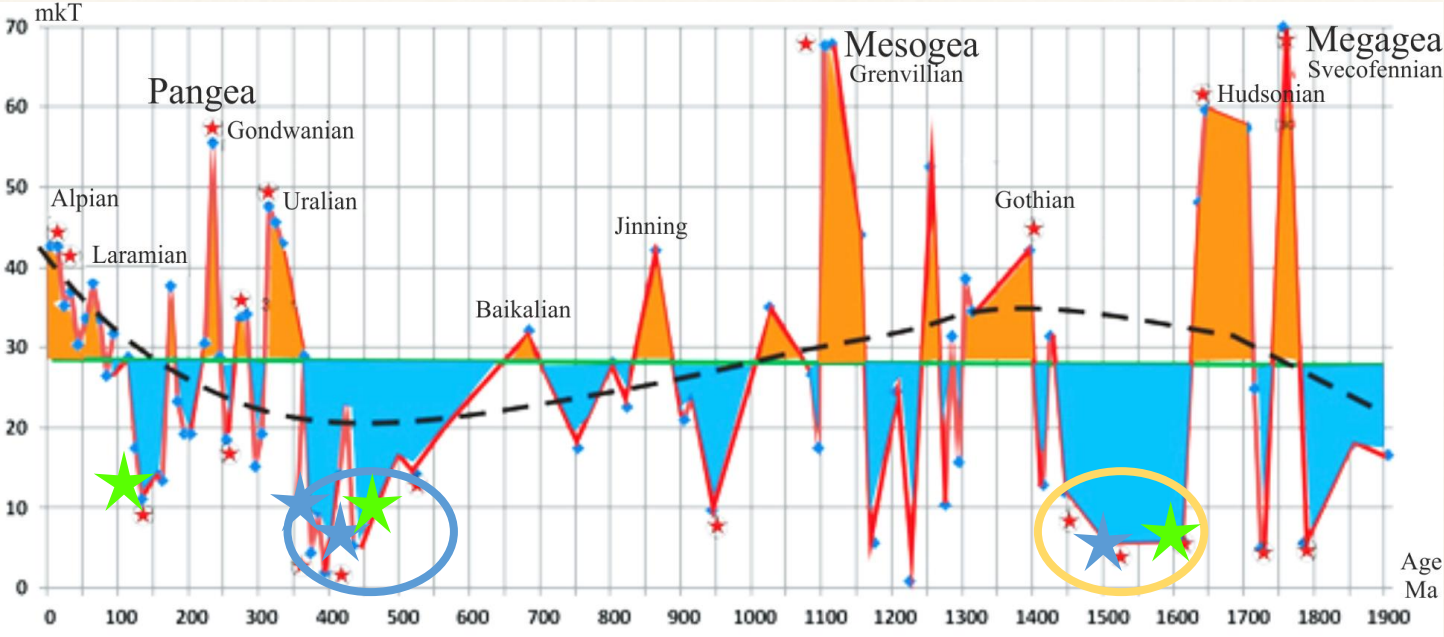
Intensity of Earth's magnetic field



(by Eduard Pinsky)

- “Oceanic” period – low magmatic activity, predominantly **sedimentary** uranium deposits
- “Continental” period –magmatic activity, predominantly **volcanic** uranium deposits
- ★ Large uranium deposits

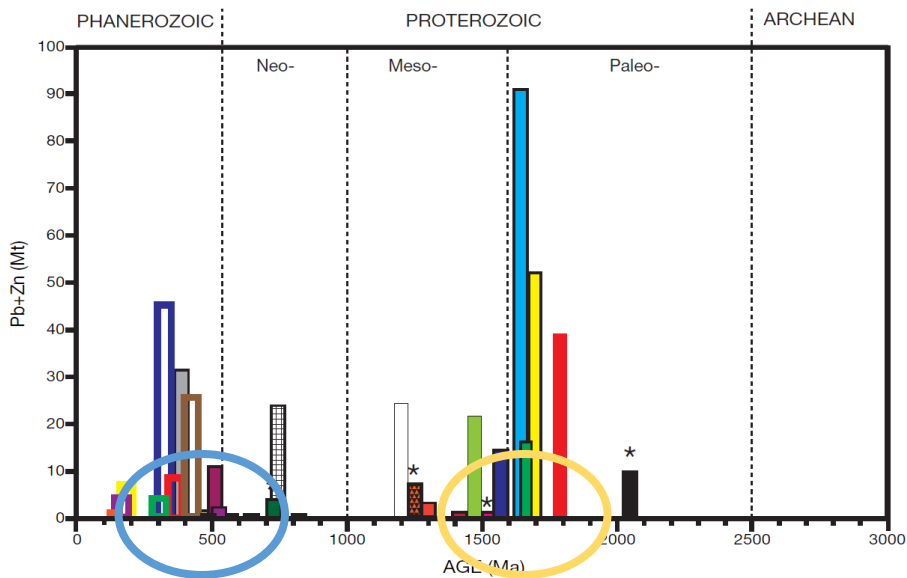
Intensity of Earth's magnetic field



- ★ MVT deposits in the database
- ★ SEDEX deposits in the database

(by Eduard Pinsky)

SEDEX and MVT deposits



- Dongshengmiao, North Park, Pinnacles
- Ambaji, Bajta Central, Ballaria, Bamnia Kalan, Baroi, Bethumni, Deri, Devpura, Ganeshpura, Gorubathan, Kankariya, Kayar, Madarpura, Mochia, Mokaupura North, Paduna North Block, Rajpura-Dariba, Rampura-Agucha, Rangpo, Rewara, Saladipura, Samodi, Sawar, Sindesar Kalan East, Sindesar Khurd, South Dedwas, Tikhi, Tiranga, Zawarmala
- Broken Hill
- Cannington, Dugald River, Pegmont
- Hilton-George Fisher, HYC, Lady Loretta, Mount Isa
- Century
- Colby, Cottonbelt, River Jordan, Ruddock Creek
- Sullivan
- Gaobanhe
- Balmat, Edwards
- Abu Samar, Cadieux, Canoas, Huogeqi, Hyatt, Jiashengpan, Kootenay King, Maramungee, Perau, Pierrepont, Tanyaokou
- Big Syncline, Black Mountain, Broken Hill, Gamsberg
- Ganesh Himal
- Kholodninskoye
- Big Ledge, Bleikvassli, Mofjellet, Rosh Pinah
- Tsongoari
- Kushk
- Angas, Duncan, H.B., Jersey, Maozu, Niujiaotang
- Dy, Faro, Grum, Mount Torrens, Swim, Vangorda
- Mel
- Citronen Fjord
- Bayindir
- Howards Pass
- Beishan, Bijlashan, Changba-Lijagou, Clear Lake, Dengjiashan, Luoba, Meggen, Qiaodongshan, Rammelsberg, Siding, Tongmugou, Triumph, Xidinggou, Yindongliang
- Arrens, Bafangshan, Cirque, Driftpile, Jason, South Cirque, Tom
- Anarraaq, Lik, Qixiashan, Red Dog, Su
- Dairi
- Filizchai
- Castellanos, Duddar, Matahambre, Santa Lucia
- Dongjiahe
- ★ Indicates poorly constrained ages

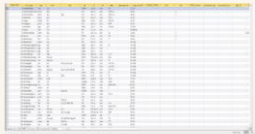
FIG. 6. Distribution of contained Pb + Zn metal in known CD deposits. Age and tonnage data for 121 deposits from Taylor et al. (2009). Deposits are identified and plotted in 20-m.y. age increments for their host rocks, or in a few cases, using the measured date of mineralization. Bins of deposits are listed from oldest to youngest in the legend.

(Leach et. al., 2010)

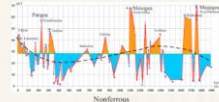
Future of metallogenic analysis



Metallogenic map



Database
>7000 deposits



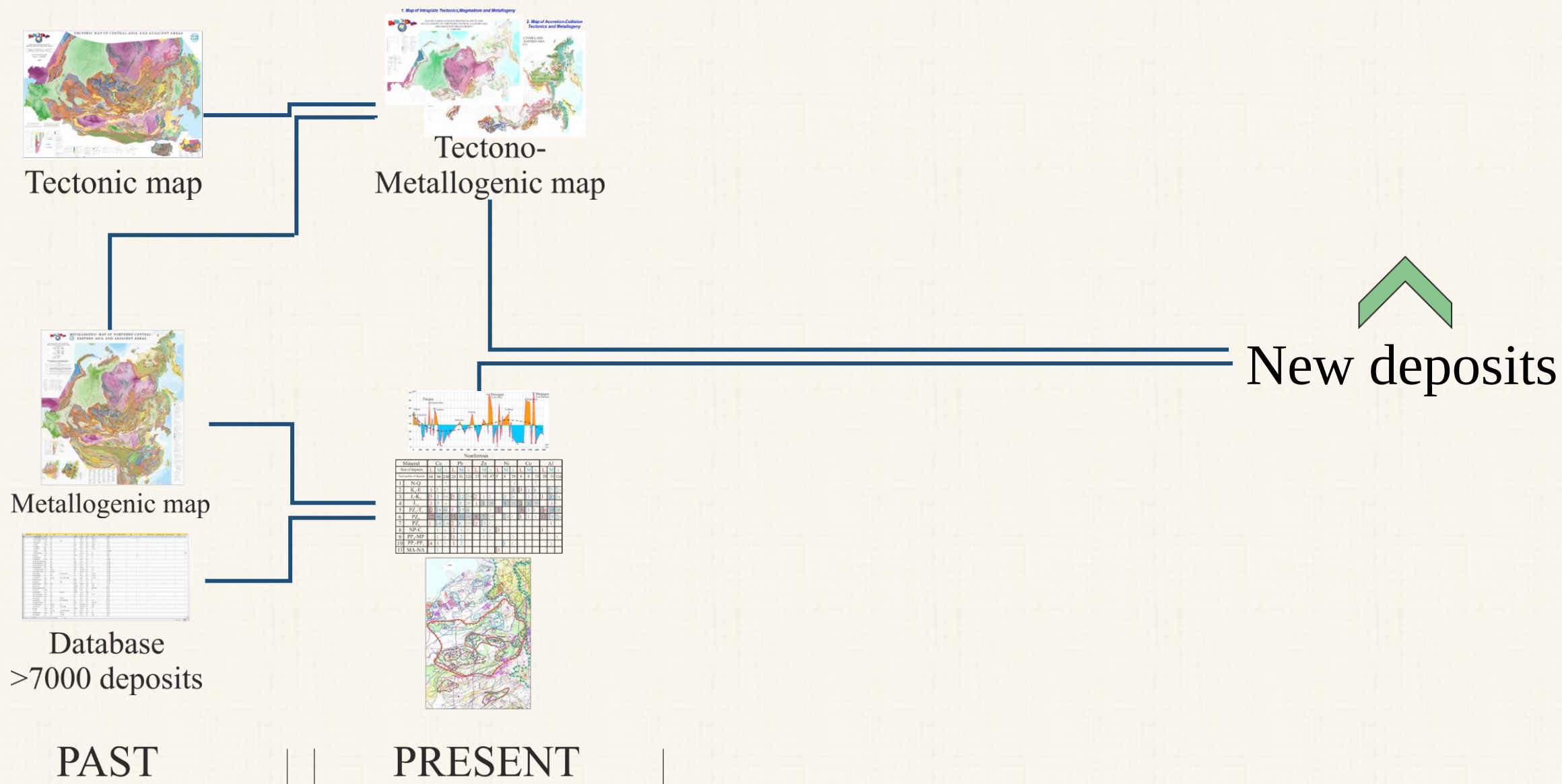
Mineral	Cu	Pb	Zn	Ni	Co	Al
1	1	1	1	1	1	1
2	1	1	1	1	1	1
3	1	1	1	1	1	1
4	1	1	1	1	1	1
5	1	1	1	1	1	1
6	1	1	1	1	1	1
7	1	1	1	1	1	1
8	1	1	1	1	1	1
9	1	1	1	1	1	1
10	1	1	1	1	1	1
11	1	1	1	1	1	1
12	1	1	1	1	1	1
13	1	1	1	1	1	1



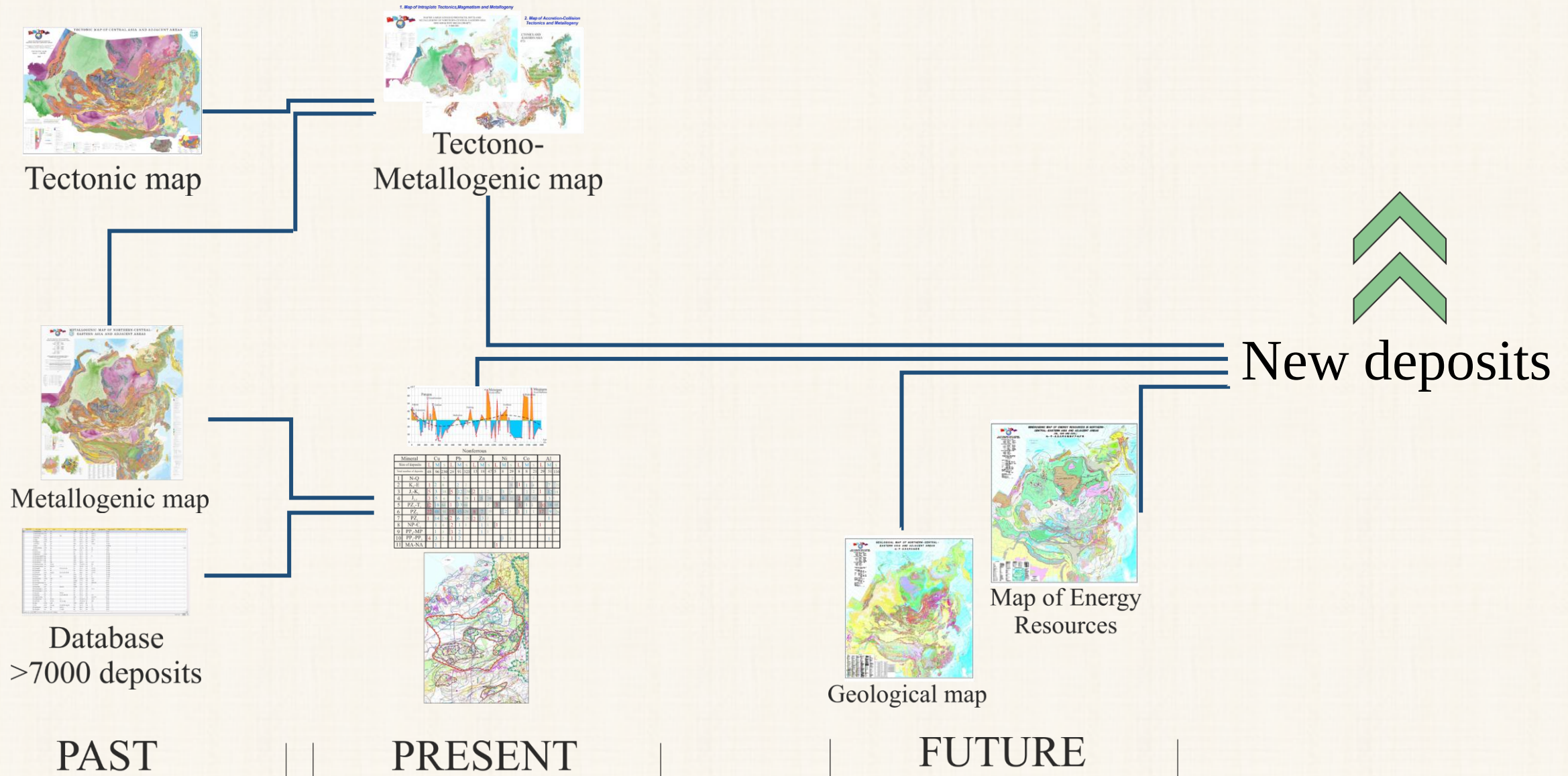
PAST

PRESENT

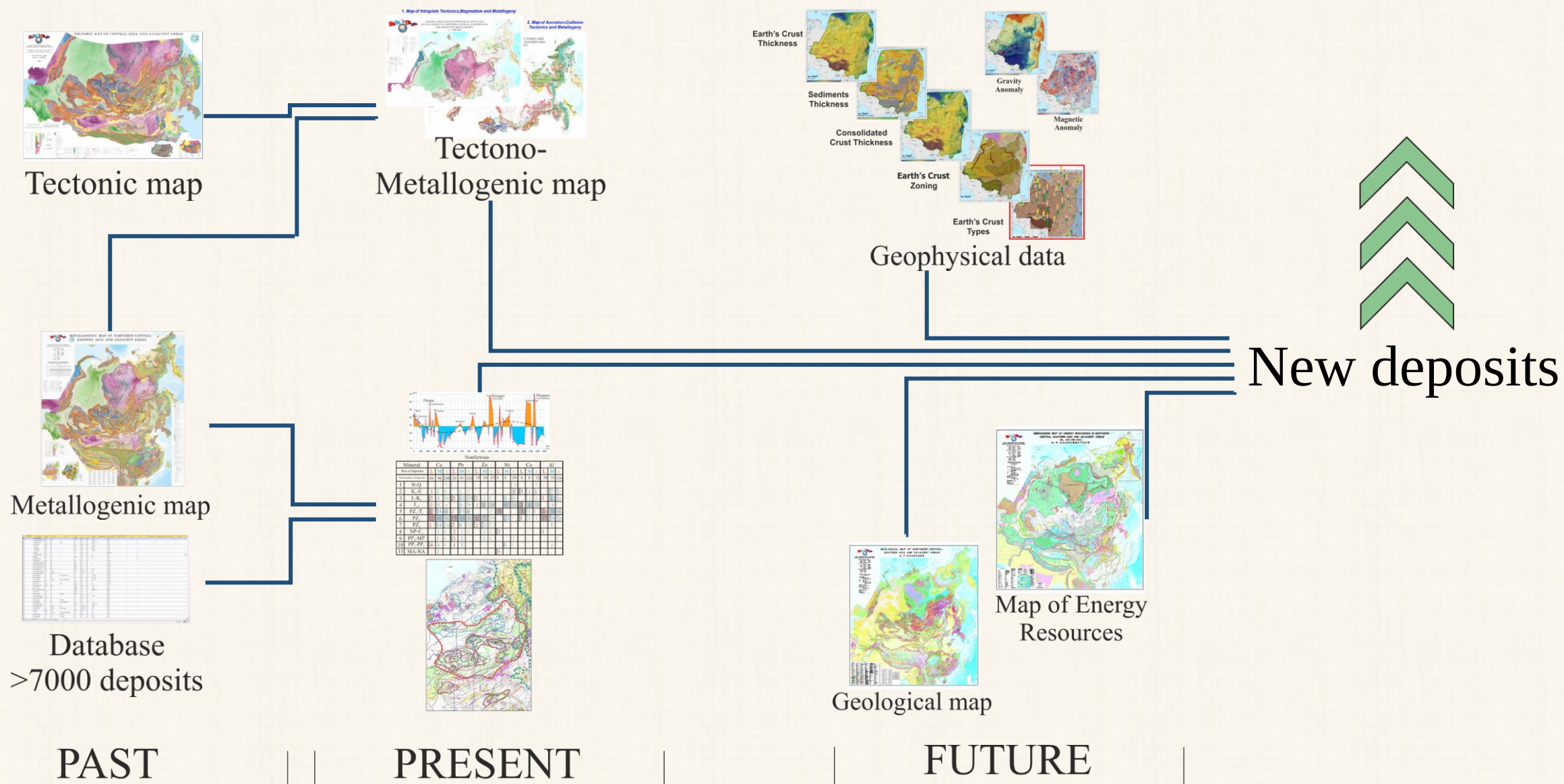
Future of metallogenic analysis



Future of metallogenic analysis



Future of metallogenic analysis



Conclusion

The next steps in the metallogenic analysis and synthesis of the project area are:

- to continue studies of the mineral resources distribution in geological time and space
- to study formation of the most important genetic types of mineral deposits of Northern-Central-Eastern Asia and its continent/ocean transition zone such as porphyry deposits, large volume deposits of noble metals in black shales etc.
- to complete the Explanatory Note for the Metallogenic map compilation and publication
- to submit published Metallogenic Map and Explanatory Note on the 36th IGC (India, 2020)

Thank you for your attention!

